Indistinguishable Single-Photon Emitters

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1 Introduction

2 Theory

all the first part will be about this paper (cite)
Pure dephasing makes solid states single photons emitters difficult outside of cryogenic temperatures. we can filter the emission to counter that but low efficiency. Current Purcell effects aren't enough. Note that dephasing makes difficult because it widens the emission spectra So we try emitter cavity coupling in the right regime for ehanced emission in the right mode. We want to find the right regime, so first is solving and plotting.
First we make following approximations (explain markovian etc) We are interested in I of course, also in beta because we want to be able to exploit. Start with beta plot because easier, as training with qutip.
explain gamma, kappa, g
add explanation of basis (why 3 dimension etc)

2.1 The Lindblad master equation

LME is useful for open systems
We initialise the system with one excitation in the emitter, $\rho_0 = |e,0\rangle\langle e,0|$, and evolve it under the Lindblad master equation:

$$\frac{d\rho}{dt} = \mathcal{L}[\rho] = -\frac{i}{\hbar}[H,\rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\left\{ L_k^\dagger L_k,\, \rho \right\} \right) \tag{1}$$

with $H = \hbar g(|e,0\rangle\langle g,1| + |g,1\rangle\langle e,0|)$ the system Hamiltonian describing the unitary aspects of the dynamics (in this case the emitter-cavity coupling), and L_k the three collapse operators encoding the dissipative channels:

- $L_1 = \sqrt{\gamma}\, |g,0\rangle\langle e,0|$ spontaneous emission (bad channel)
- $L_2 = \sqrt{\kappa}\, |g,0\rangle\langle g,1|$ cavity decay into (good channel)
- $L_3 = \sqrt{\gamma^*}\, |e,0\rangle\langle e,0|$ pure dephasing (no population change)

3 Efficiency

The efficiency β is defined as the probability that a photon, following the initial excitation of the emitter, is emitted through the cavity's good channel rather than lost to the environment. Formally, it is the total flux of photons leaving through the cavity mode:

$$\beta = \kappa \int_0^\infty \langle \hat{n}_{\text{cav}}(t) \rangle \, dt = \kappa \int_0^\infty \text{Tr}[P_{\text{cav}}\, \rho(t)] \, dt \tag{2}$$

where $P_{\text{cav}} = |g,1\rangle\langle g,1|$ is the projector onto the cavity-occupied state and κ is the cavity decay rate. Physically, this makes a lot of sense as it is intuitively the average number of photons in the cavity over time, weighted by κ, the rate at which each photon escapes through the good channel.

3.1 Numerical approach

describe the QuTiP simulation, convergence study, and plots
attempted solutions, mesh etc

Failure of the numerical approach

The numerical simulation using QuTiP's master-equation solver (`mesolve`) failed to converge for a significant portion of the (g, κ) parameter grid. The root cause is *stiffness*: the system involves timescales that differ by many orders of magnitude. Specifically, the pure dephasing rate $\gamma^* = 10^4\,\gamma$ is much larger than γ, while at the same time large values of g or κ introduce respectively rapid Rabi oscillations or fast cavity decay. Resolving the fastest timescale while integrating over the slowest one demands an extremely large number of time steps, making the simulation computationally intractable for the upper-right corner of the parameter space. This observation was the motivation to start looking for an analytical solution to the problem.

check the timescale discussion
explain the attempts at having lower tolerance for both, fix nt,

3.2 Analytical approach

3.2.1 Vectorisation of the density matrix

explain the logic of vectorisation and motivation (linear) + cite the ref in code
The density matrix can be flattened into a column vector by stacking its columns:

$$\rho \longmapsto |\rho\rangle\rangle \in \mathbb{C}^{d^2} \tag{3}$$

3x3 to 1x9 - expliquer dimensions?
This turns the Lindblad equation into an ordinary linear ODE:

$$\frac{d}{dt}|\rho\rangle\rangle = \hat{\mathcal{L}}\,|\rho\rangle\rangle \implies |\rho(t)\rangle\rangle = e^{\hat{\mathcal{L}}t}\,|\rho_0\rangle\rangle \tag{4}$$

$\hat{\mathcal{L}}$ is a 9×9 complex matrix called the *Liouvillian superoperator*. Using the identity $|AXB\rangle\rangle = (B^\top \otimes A)\,|X\rangle\rangle$, it takes the explicit form:

$$\hat{\mathcal{L}} = -\frac{i}{\hbar}(I \otimes H - H^\top \otimes I) + \sum_k \left[L_k^* \otimes L_k - \frac{1}{2}\Big(I \otimes L_k^\dagger L_k + (L_k^\dagger L_k)^\top \otimes I\Big)\right] \tag{5}$$

Expectation values then become inner products in the vectorised space:

$$\mathrm{Tr}[P_{\text{cav}}\,\rho(t)] = \langle\langle P_{\text{cav}}\,|\rho(t)\rangle\rangle \tag{6}$$

so that the efficiency (2) reads:

$$\beta = \kappa \int_0^\infty \langle\langle P_{\text{cav}}|e^{\hat{\mathcal{L}}t}|\rho_0\rangle\rangle\,dt \tag{7}$$

3.2.2 The singular Liouvillian

All eigenvalues of $\hat{\mathcal{L}}$ have non-positive real parts, and there is a unique zero eigenvalue corresponding to the steady state $\rho_{\text{ss}} = |g,0\rangle\langle g,0|$ (all excitation has decayed). This makes $\hat{\mathcal{L}}$ singular, so one cannot directly invert the integral in (7).
explain standard fix: substract steady state
$\sigma(t) = \rho(t) - \rho_{\text{ss}}$
decays to 0 + see hand notes for the other steps
$\frac{d}{dt}|\sigma\rangle\rangle = \hat{\mathcal{L}}\,|\sigma\rangle\rangle$. Integrating both sides from 0 to ∞:

$$\hat{\mathcal{L}}\int_0^\infty |\sigma(t)\rangle\rangle\,dt = \left[e^{\hat{\mathcal{L}}t}\,|\sigma_0\rangle\rangle\right]_0^\infty = -\,|\sigma_0\rangle\rangle \tag{8}$$

where $|\sigma_0\rangle\rangle = |\rho_0 - \rho_{ss}\rangle\rangle$.
This is now invertible on the subspace orthogonal to the null space of L?

$$\int_0^\infty |\sigma(t)\rangle\rangle \, dt = -\hat{\mathcal{L}}^{-1} \, |\sigma_0\rangle\rangle \tag{9}$$

Since $\mathrm{Tr}[P_{\mathrm{cav}} \, \rho_{\mathrm{ss}}] = 0$ (there are no photons in the cavity at steady state), we have

$$\int_0^\infty \langle\langle P_{\mathrm{cav}} | \rho(t)\rangle\rangle \, dt = \int_0^\infty \langle\langle P_{\mathrm{cav}} | \sigma(t)\rangle\rangle \, dt \tag{10}$$

and the efficiency reduces to the closed-form expression:

$$\boxed{\beta = -\kappa \, \langle\langle P_{\mathrm{cav}} | \hat{\mathcal{L}}^{-1} | \rho_0 - \rho_{\mathrm{ss}}\rangle\rangle} \tag{11}$$

In practice this reduces to a single 9×9 linear solve $\hat{\mathcal{L}} \, x = - |\rho_0 - \rho_{\mathrm{ss}}\rangle\rangle$, which is exact and can be computed numerically very fast, and importantly independently of the values of g, κ, or γ^*.

3.3 Results

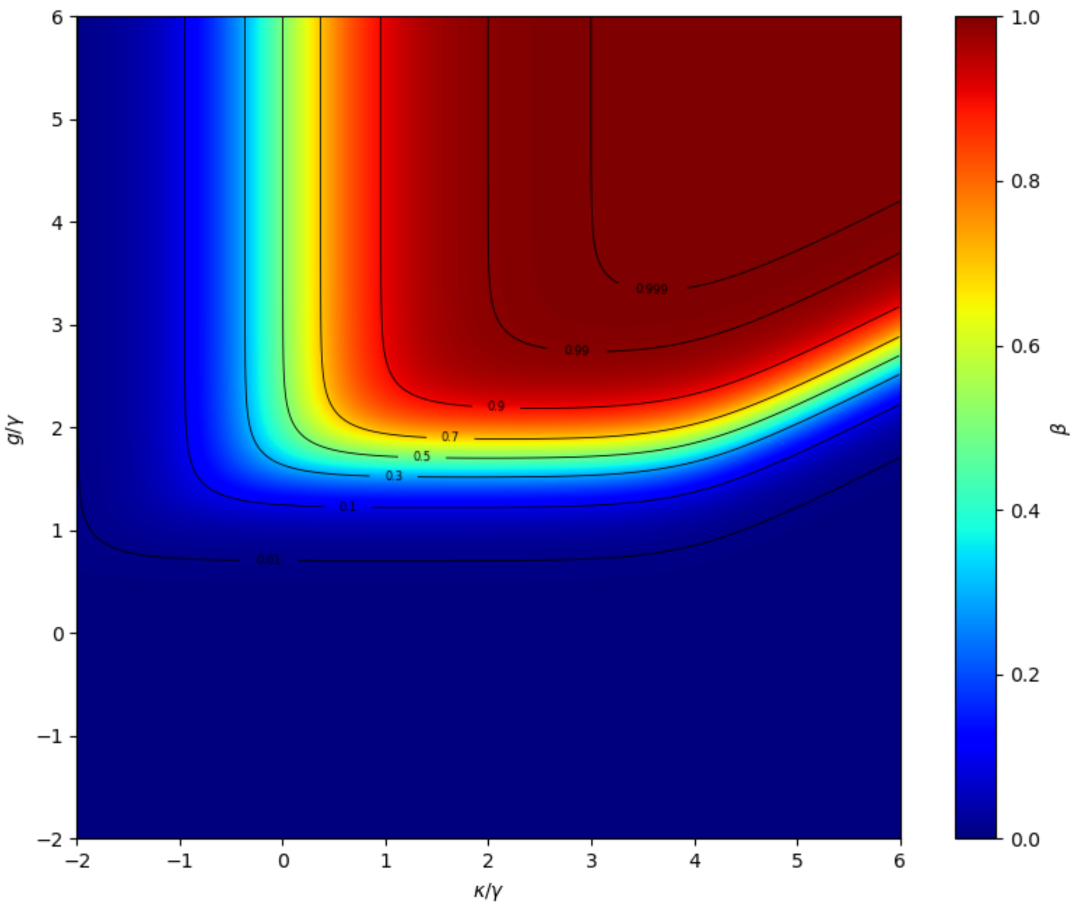

Figure 1: Efficiency β as a function of the coupling g and cavity decay rate κ, for a fixed ratio $\gamma^*/\gamma = 10^4$.

3.4 Remarks

In hindsight, it was possible to solve this more easily, wihtout the vectorisation. Howerever, the formalism is useful for the part on indistinguishability so we accept to pay that price, and it is

already done anyways.
Still, we can sketch the approach, as a system of ODE.
In the basis $\{|g,0\rangle, |g,1\rangle, |e,0\rangle\}$, ρ is:

$$\rho = \begin{pmatrix} \rho_{gg} & \rho_{gc} & \rho_{ge} \\ \rho_{cg} & \rho_{cc} & \rho_{ce} \\ \rho_{eg} & \rho_{ec} & \rho_{ee} \end{pmatrix} \tag{12}$$

Each element evolves as $\dot{\rho}_{ij} = \langle i|\dot{\rho}|j\rangle$, given by the Lindblad equation (1).
First term: The commutator $[H, \rho]$ gives for each element:

$$\langle i|[H,\rho]|j\rangle = \langle i|H\rho|j\rangle - \langle i|\rho H|j\rangle \tag{13}$$

with, as a reminder, the Hamiltonian $H = \hbar g(|e,0\rangle\langle g,1| + |g,1\rangle\langle e,0|)$.
For example for $\dot{\rho}_{ee}$:

$$\begin{aligned} -\frac{i}{\hbar}\langle e,0|[H,\rho]|e,0\rangle &= -\frac{i}{\hbar}\Big(\langle e,0|H\rho|e,0\rangle - \langle e,0|\rho H|e,0\rangle\Big) \\ &= -\frac{i}{\hbar}\left(\sum_k \langle e,0|H|k\rangle\langle k|\rho|e,0\rangle - \sum_l \langle e,0|\rho|l\rangle\langle l|H|e,0\rangle\right) \\ &= -ig\left(\rho_{ce} - \rho_{ec}\right) \end{aligned}$$

Most terms vanish before the last step.
After some tedious computation that can be found in Appendix, we finally get

$$\begin{aligned} -\frac{i}{\hbar}[H,\rho]_{ee} &= -ig(\rho_{ce} - \rho_{ec}) \\ -\frac{i}{\hbar}[H,\rho]_{cc} &= +ig(\rho_{ce} - \rho_{ec}) \\ -\frac{i}{\hbar}[H,\rho]_{ec} &= +ig(\rho_{ee} - \rho_{cc}) \end{aligned}$$

ajouter le reste en appendix sauf si trop chiant, et mettre la ref
Second term: The contribution of each operator L_k is:

$$L_k\rho L_k^\dagger - \frac{1}{2}(L_k^\dagger L_k\rho + \rho L_k^\dagger L_k)$$

Projected onto element (i,j):

$$\langle i|L_k\rho L_k^\dagger|j\rangle - \frac{1}{2}(\langle i|L_k^\dagger L_k\rho|j\rangle + \langle i|\rho L_k^\dagger L_k|j\rangle)$$

Take $L_1 = \sqrt{\gamma}\,|g,0\rangle\langle e,0|$ as an example.
First, $L_1^\dagger L_1 = \gamma\,|e,0\rangle\langle e,0| = \gamma P_e$, so the anticommutator term gives:

$$-\frac{\gamma}{2}\langle i|\{P_e,\rho\}|j\rangle = -\frac{\gamma}{2}(\langle i|P_e\rho|j\rangle + \langle i|\rho P_e|j\rangle)$$

And the $L_1\rho L_1^\dagger$ term gives:

$$\gamma\langle i|g,0\rangle\langle e,0|\rho|e,0\rangle\langle g,0|j\rangle = \gamma\,\delta_{i,g_0}\delta_{j,g_0}\,\rho_{ee}$$

So L_1 drains ρ_{ee} into ρ_{gg} at rate γ, and kills all coherences involving $|e,0\rangle$ at rate $\gamma/2$.
The same procedure is applied to $L_2 = \sqrt{\kappa}\,|g,0\rangle\langle g,1|$ and $L_3 = \sqrt{\gamma^*}\,|e,0\rangle\langle e,0|$.

si ça a un interet mettre dans annexe aussi le reste mais bon
Now, applying the dissipator contributions to each equations yields:

$$\begin{aligned}\dot{\rho}_{ee} &= -\gamma\,\rho_{ee} - ig(\rho_{ce} - \rho_{ec}) \\ \dot{\rho}_{cc} &= -\kappa\,\rho_{cc} + ig(\rho_{ce} - \rho_{ec}) \\ \dot{\rho}_{ec} &= -\frac{\gamma + \kappa + \gamma^*}{2}\,\rho_{ec} + ig(\rho_{ee} - \rho_{cc})\end{aligned}$$

By hermiticity, $\rho_{ce} = \rho_{ec}^*$, so $\rho_{ce} - \rho_{ec} = -2i\,\mathrm{Im}(\rho_{ec})$. Defining $v = \mathrm{Im}(\rho_{ec})$, the system simplifies to three real equations:

$$\boxed{\begin{aligned}\dot{\rho}_{ee} &= -\gamma\,\rho_{ee} - 2g\,v \\ \dot{\rho}_{cc} &= -\kappa\,\rho_{cc} + 2g\,v \\ \dot{v} &= g(\rho_{ee} - \rho_{cc}) - \frac{\gamma + \kappa + \gamma^*}{2}\,v\end{aligned}}$$

This is $\dot{\mathbf{x}} = M\mathbf{x}$ with $\mathbf{x} = (\rho_{ee}, \rho_{cc}, v)^\top$ and

$$M = \begin{pmatrix} -\gamma & 0 & -2g \\ 0 & -\kappa & 2g \\ g & -g & -\dfrac{\gamma + \kappa + \gamma^*}{2} \end{pmatrix}$$

and initial condition $\mathbf{x}(0) = (1, 0, 0)^\top$. To get β, define $\mathbf{X} = \int_0^\infty \mathbf{x}(t)\,dt$. Integrating $\dot{\mathbf{x}} = M\mathbf{x}$ directly gives $M\mathbf{X} = -\mathbf{x}(0)$, since $\mathbf{x}(\infty) = 0$. M is invertible because $|g, 0\rangle$ is not tracked here, so no zero eigenvalue appears.
Solving this system gives is simple numerically, we get $X_{cc} = \int_0^\infty \rho_{cc}(t)\,dt$ and

$$\beta = \kappa\,X_{cc} \tag{14}$$

4 Indistinguishability

5 Approximations

6 Geometries

7 Conclusion

testing